Veriscrypt AI

Confidential Computing at Scale

Hardware-Accelerated Privacy for the GenAI Era

The AI Privacy Crisis

Why Enterprise AI is currently stalled

Privacy vs. Utility Trade-off

Data Vulnerability

Public cloud LLMs require data to be decrypted for processing. This exposes proprietary corporate IP and sensitive PII to cloud providers and potential breaches.

Compliance Friction

HIPAA, GDPR, and CCPA mandates prevent regulated industries (Health, Finance, Gov) from utilizing standard cloud-hosted AI endpoints for sensitive workloads.

The Performance Wall

Software FHE Limits

Fully Homomorphic Encryption (FHE) is the holy grail of privacy, allowing computation on encrypted data. However:

- Software-based FHE is 10,000x slower than standard inference.
- Bootstrapping bottlenecks prevent real-time scaling.
- Prohibitive costs for production-grade Transformers.

Introducing VerisCore-1

The World's First ASIC for Private AI

Silicon-Level Privacy

Native Acceleration

VerisCore-1 is a dedicated ASIC co-processor tailored for R-LWE primitives and Transformer attention mechanisms.

By mapping the Cheon-Kim-Kim-Song (CKKS) scheme directly into hardware logic, we eliminate the traditional 10,000x overhead penalty found in software.

Technological Breakthroughs

0% Overhead

Our hardware pipeline achieves a 1:1 speed parity with unencrypted FP16 inference for 70B parameter models.

CKKS Silicon NTT

Native hardware support for the CKKS scheme and Number Theoretic Transform (NTT) modular arithmetic.

Zero-Delay Bootstrap

A proprietary on-chip bootstrapping pipeline refreshes ciphertexts in 4.2μs, enabling continuous inference.

Performance Benchmarks

Metric	Software FHE	VerisCore-1	Baseline (NVIDIA H100)
Latency (70B Model)	4,200 sec	0.12 sec	0.12 sec
Throughput (Tokens/s)	0.001	45.0	45.0
Execution Speed Parity	10,000x Penalty	1:1 Parity	Reference
Crypto Overhead	Prohibitive	Zero-Delay	N/A

Traction & Pilot Milestones

64

PCIE CARDS SHIPPED

Enterprise Momentum

Veriscrypt is actively shipping VerisCore-1 PCIe Gen5 accelerator cards to Tier-1 Cloud Service Providers.

- Taped out on TSMC 4nm process node (Q1 2026).
- Successful integration with Llama-3 and Mixtral.
- Active bare-metal pilots bypassing Nitro layer for speed.

Deployment Infrastructure

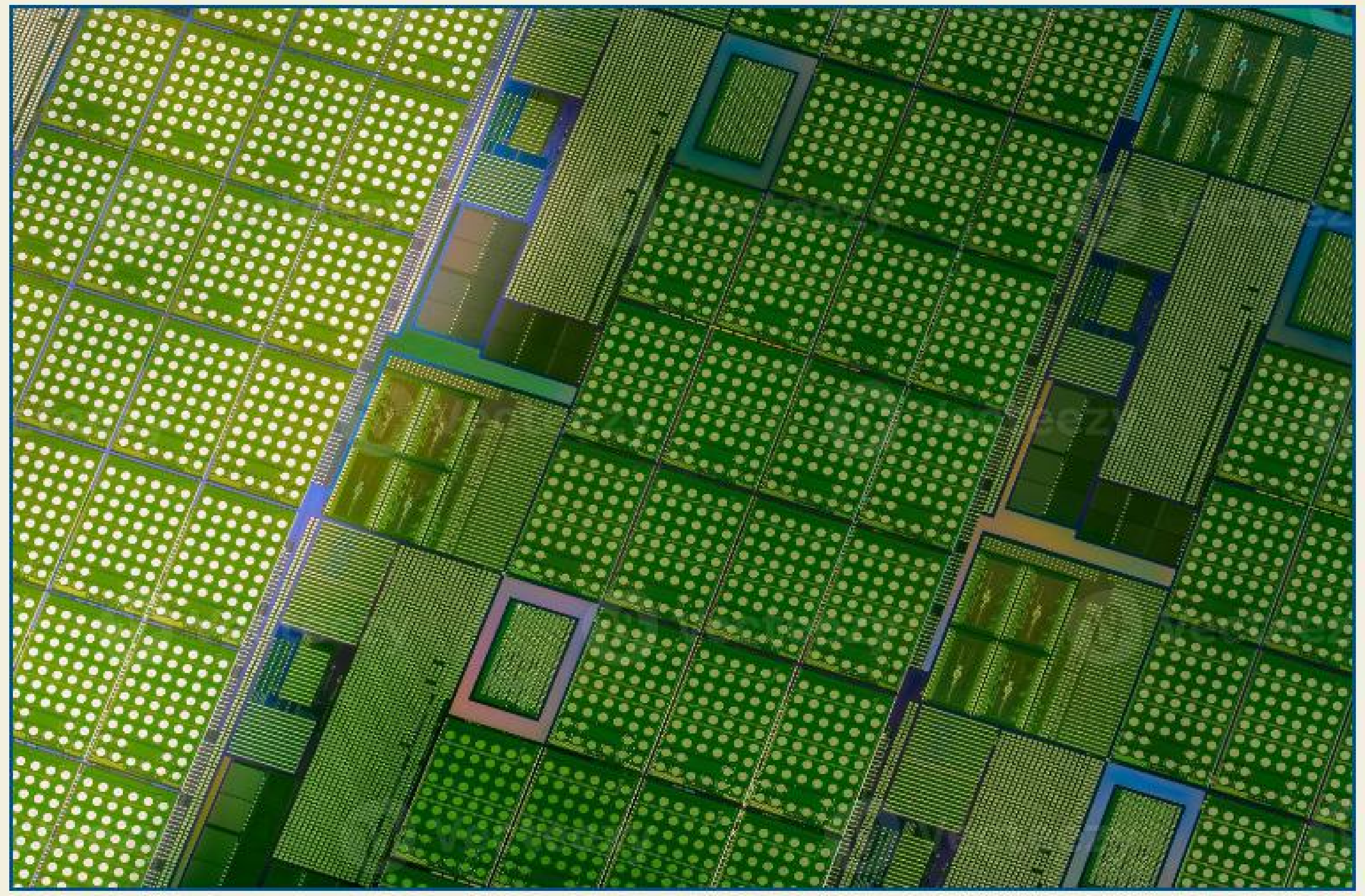
AWS Deployment

AWS us-west-2

64-card bare-metal cluster for health-tech design partners.

Azure East US 2

Private evaluation sandbox for government cloud pilots.



TSMC Foundry

Scaling VerisCore-2 production to meet Seed round demand.

Vision for 2026

"Veriscrypt AI is turning the cryptographic holy grail into a commodity. We aren't just building a chip; we are building the foundation for a truly private internet."

— Dr. Alex Chen, CEO & Co-Founder

Questions?

Contact: alex@veriscrypt.ai | www.veriscrypt.ai

$5M Seed Round | Series A Q1 2027

Image Sources

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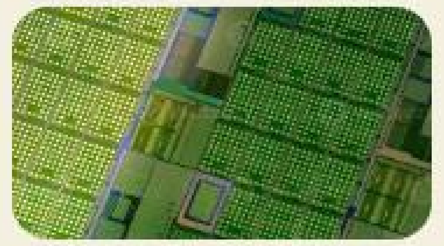

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